

Sourcing the data behind the APIs

`main` exposes 42 endpoints, but almost none of them fetch anything. The whole 21k-line `src/` tree makes **one** outbound HTTP call. Everything else — MCA, GST, Udyam, eCourts, bureau scores, telephony, WhatsApp — resolves from the local database or returns a hardcoded literal.

This document is about closing that gap: for each capability the product promises, where the real data comes from, what it costs you in eligibility rather than money, and how to wire a real provider into the existing adapter seam.

Commercial terms in this space are negotiated, change often, and are usually behind a sales conversation. Nothing here quotes a price. Treat every provider as "confirm current terms directly" — the durable part of this document is the *eligibility* and *architecture*, which move far more slowly than rate cards.

1. What actually has to be sourced

`src/lib/verification-gateway/types.ts` defines 17 report types. Grouped by where the data must come from:

GROUP	REPORT TYPES	REAL SOURCE
Corporate registry	<code>director_details</code> , company master data	MCA21
Tax registration	<code>gst_slab_check</code> , <code>gst_exact_turnover</code> , <code>gst_supreme_report</code>	GSTN
MSME status	<code>msme_report</code>	Udyam portal
Trade	<code>import_export_report</code>	DGFT (IEC)
Identity	<code>mobile_to_pan</code> , <code>pan_to_mobile_email</code> , <code>mobile_identity</code> , <code>mobile_to_address</code> , <code>address_enrichment</code> , <code>find_someone</code>	Aggregators over NSDL/Protean, telco KYC
Litigation	<code>court_case_history</code> , <code>fir_check</code>	eCourts, NCLT, state police portals
Credit	<code>bureau_report</code> , <code>payment_behaviour</code>	CIBIL / Experian / CRIF / Equifax
Composite	<code>company_supreme_report</code>	Aggregator bundle of the above

Plus the outbound channels: voice, WhatsApp, SMS, email, e-sign.

2. Three gates you hit before any pricing conversation

These decide what you are *allowed* to buy, and they are the real constraint.

Credit bureau data is licensed, not sold. Under the Credit Information Companies (Regulation) Act 2005 (CICRA), only entities registered with the RBI as a Credit Institution — a bank, NBFC, housing finance company, or a notified Specified User — or entities registered with SEBI or IRDAI, can pull bureau data. You cannot sign up on a website for this. Bureaus also require a data-security audit before granting direct access. There are three realistic paths: become a Credit Institution, partner with one and pull under their membership under a written agreement (common for early-stage lending fintechs), or buy a derived product from an aggregator that holds the licence.

SMS requires TRAI DLT registration. Three phases, all mandatory before a single message sends: register the entity, register the sender ID (header), register each template. Unregistered traffic is blocked by the operators, not merely rate-limited. WhatsApp does **not** need DLT — it is not classified as telecom SMS — but does need a Business Service Provider account approved by Meta.

Consent is a legal artefact under the DPDP Act 2023. For a recovery product this is not paperwork: you are processing a third party's personal data to contact them about a debt. The consent record, its provenance, and its withdrawal path need to exist before the first call, not after.

3. Tier 1 — free government sources

Enough to build a real V0 without a single commercial contract.

SOURCE	WHAT YOU GET	ACCESS
MCA21 V3	Company/LLP master data, CIN, status, directors, charges	Public portal, free. Basic master data needs no login; filing history and charge data do. No documented public API — this is why <code>main</code> 's MCA adapter is manual-entry.
GST portal	GSTIN validity, legal name, status, registration date, taxpayer type	Public search, no credentials. Turnover is not exposed — that is why the product asks for GSTR-3B uploads.
Udyam	MSME registration status, enterprise classification	Public verification, no auth
eCourts / NCLT	Case history, insolvency proceedings	Public search; no bulk API, CAPTCHA-gated
API Setu	MeitY's open API platform — validates GSTIN, IEC and PAN registration details	Free tier; register for a key

API Setu is the one government route with a genuine API, and it is already the single real integration in `main`:

```
// src/lib/verification-gateway/clients.ts
const res = await fetch(`https://apisetu.gov.in/gst/v1/turnover/${gstIn}`, {
  headers: { "X-API-KEY": process.env.APISETU_API_KEY },
});
```

The honest limit of Tier 1: no bulk access, CAPTCHAs, and no SLA. It is correct for manual, operator-driven verification — exactly what `main` built — and wrong for anything automated at volume.

WHY `MAIN` CHOSE MANUAL ENTRY, AND WHY THAT WAS RIGHT

Its adapters say so explicitly:

```
/**
 * MCA Adapter - V0 Public-Data Mode
 * There is no free, unauthenticated public JSON API for MCA21.
 * ...
 * Future: swap openMcaPortalUrl() + parseManualPayload() with an
 * authorized API Setu / Karza / Digitap call - the interface is the same.
 */
```

An operator opens the portal, reads the page, types what they saw, and the server files it as `sourceStatus = "USER_PROVIDED"`. That is a defensible V0: the provenance is tracked rather than fabricated, and the interface is already shaped for a real provider to drop in.

4. Tier 2 — commercial aggregators

One contract covering identity, business and litigation data. This is the practical answer for most of the 17 report types.

PROVIDER	NOTES
Perfios (Karza)	Karza was acquired by Perfios in February 2022. Named directly in <code>main</code> 's adapter comments. Its TotalKYC covers identity, business, financial and legal data; one of the deepest Indian data libraries. Sits alongside Perfios' bank-statement analysis.
Signzy	KYC/KYB, video KYC, bank verification
IDfy	Identity, background verification, litigation checks
Digitap	Named in <code>main</code> 's comments; identity and alternate data
Surepass, Zoop, Setu	Lighter-weight, faster to onboard, often better documented for small teams

How to choose. Coverage per report type matters more than headline price — ask for a per-endpoint coverage matrix against the 17 types above and check the *fill rate*, not just whether an endpoint exists. Then check: pricing model (per-

hit vs subscription vs committed volume), whether failed lookups are billed, sandbox quality, and contractual data-retention terms, which matter under DPDP.

Onboarding is typically a KYC of your own company, an agreement, and a security review — days to weeks, not minutes.

5. Tier 3 — credit bureaus

The hardest and most valuable tier. Four bureaus operate in India: **TransUnion CIBIL**, **Experian**, **CRIF High Mark**, **Equifax**. `main` declares a key for the first three.

Commercial (business) bureau reports are a distinct product from consumer reports and are what a B2B credit-recovery product actually needs — CIBIL's commercial product covers entity credit history, not an individual's score.

Access is gated by \$2 above. Realistically:

- 1. **Direct membership** — only if you are already an RBI-registered Credit Institution.
- 2. **Under a partner's membership** — via co-lending or a written data-sharing agreement.
- 3. **Through an aggregator** that holds the licence and resells a derived product. Fastest, least control, and check carefully what you are contractually allowed to *display* to your own users.

Budget the eligibility work, not the integration. The API itself is straightforward; becoming allowed to call it is the project.

6. Outbound channels

CHANNEL	WHAT <code>MAIN</code> DECLARES	WHAT IT ACTUALLY NEEDS
Voice	<code>VOBIZ_SIP_ENDPOINT</code> , Asterisk	A SIP/telephony provider (Exotel, Knowlarity, Twilio, or a direct SIP trunk). <code>main</code> writes a <code>Call</code> row without dialling.
WhatsApp	<code>WHATSAPP_BSP_API_KEY</code>	A Meta-approved BSP (Gupshup, Twilio, and others). Template approval required. No DLT.
SMS	via BSP	TRAI DLT registration first — entity, header, template. Non-negotiable.
Email	<code>SENDGRID_API_KEY</code>	SendGrid or equivalent; domain auth (SPF/DKIM) before deliverability is usable.
TTS	<code>POLLY_ACCESS_KEY</code>	AWS Polly. Indian-language neural voices exist and are the reason the product ships scripts in four languages.
e-Sign	—	An ASP/ESP such as Digio or Leegality over NSDL/CDSL eSign, for Aadhaar OTP signatures under IT Act §3A.

Two behaviours in `main` to fix before any of this is real, both already surfaced in the Android UI:

- The **direct voice action bypasses the TRAI calling-window and frequency-cap checks** that the policy tick applies. Under a real telephony contract that is a compliance breach, not a bug in a demo.
- `/api/recovery/settle` is not idempotent — a replayed webhook applies twice. Fix before money moves through it.

7. What `main` declares vs what it needs

Its `.env.example` lists thirteen keys:

```
DATABASE_URL  REDIS_URL          APISETU_API_KEY  CIBIL_API_KEY
EXPERIAN_API_KEY  CRIF_API_KEY  WHATSAPP_BSP_API_KEY
SENDGRID_API_KEY  VOBIZ_SIP_ENDPOINT  POLLY_ACCESS_KEY
GEMINI_API_KEY    IT_PORTAL_API_KEY  GST_PORTAL_API_KEY
```

Only `DATABASE_URL` is required to run. Five of these are read by no code at all. The rest are placeholders for integrations that do not exist yet — useful as a statement of intent, misleading if read as a dependency list.

8. A phased sourcing plan

Phase 0 — free, today. Register for an API Setu key. Keep MCA/Udyam/eCourts on the existing manual-entry flow with `USER_PROVIDED` provenance. This is genuinely shippable for a small operator team and costs nothing.

Phase 1 — one aggregator. Contract a single Tier 2 provider covering identity, business and litigation. Swap the four V0 adapters behind their existing interface. This is where automated verification actually begins.

Phase 2 — communications. DLT registration (start early; it gates SMS entirely), a WhatsApp BSP, SendGrid domain auth, and a telephony provider. Fix the calling-window bypass before the first real call.

Phase 3 — bureau. Begin the eligibility work in parallel with Phase 1, because it is the long pole. Until then, take bureau-derived data through the aggregator.

Throughout — consent. Build the DPDP consent record before Phase 2, not after.

9. Wiring a real provider in

The adapter seam already exists and is the best-built part of `main`. `VerificationAdapter` is a typed contract with a registry, a DB-backed TTL cache keyed on `(subjectType, subjectId, reportType)`, and parallel fan-out with per-report degradation:

```
export interface VerificationAdapter {  
  provider: string;  
  supportedReports: ReportType[];  
  getReport(subject, reportType): Promise<NormalizedReport>;  
}
```

To add a real provider:

1. Implement `VerificationAdapter` against the vendor SDK.
2. Register it in `src/lib/verification-gateway/adapters.ts`.
3. Keep `sourceStatus` honest — emit `VERIFIED` only for data a provider actually returned, never for manual entry.
4. Leave the V0 manual path in place as the fallback when a lookup fails or is not covered.

Nothing in the Android client changes. It consumes `NormalizedReport` through `backgroundcheck/data/BackgroundCheckApi.kt`, so a provider swap is invisible to it — which is the point of the seam.

One thing the client will need updating for: it currently labels manual data as user-provided and states plainly that certain checks are not performed. **Those notices are tied to real behaviour.** When a provider makes a check real, update the copy — and only then.

Sources

- [API Setu](#) — Government of India open API platform
- [API Setu documentation: use cases](#)
- [MCA21 V3 portal](#)
- [Perfios](#) — KYC/KYB products (Karza)
- [Identity verification API vendor guide, 2026](#)
- [Credit bureau API integration overview](#)
- [TransUnion CIBIL access paths for lending teams](#)
- [DLT registration: TRAI guide for India](#)
- [India SMS regulations and TRAI compliance](#)
- [WhatsApp Business API in India: BSP, DPDP, integration](#)